

Tax Shield Intensity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Tax Shield Intensity (TSI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on TSI achieves an annualized gross (net) Sharpe ratio of 0.55 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 48 (46) bps/month with a t-statistic of 4.65 (4.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price) is 29 bps/month with a t-statistic of 2.90.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this information incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence documents systematic delays in how markets process and respond to value-relevant signals. Tax-related information presents a particularly compelling setting to examine these informational frictions. Despite the economic significance of corporate tax shields for firm valuation, the complexity and opacity of tax reporting create natural barriers to rapid information diffusion. Tax Shield Intensity (TSI), which captures the relative magnitude of tax benefits from debt financing and other deductible expenses, represents a fundamental yet underappreciated determinant of firm value that may be subject to delayed market recognition.

The slow diffusion of tax-related information creates predictable patterns in stock returns through investor underreaction to TSI signals. Following the limited attention framework of [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), investors face cognitive constraints that prevent them from fully processing all available information simultaneously. Tax information proves particularly susceptible to these attention constraints due to its technical complexity and placement in financial statement footnotes rather than headline earnings figures. The gradual information diffusion model of [Hong et al. \(2000\)](#) predicts that value-relevant signals spread slowly across the investor population, generating momentum in returns as information cascades from sophisticated to less sophisticated market participants. For TSI specifically, firms with high tax shield intensity possess valuable tax benefits that reduce their effective cost of capital, yet these benefits require careful analysis of tax footnotes and reconciliation schedules to properly assess. The information processing costs associated with extracting and interpreting TSI data create natural frictions in

price discovery. Consistent with [Peng and Xiong \(2006\)](#), investors optimally allocate limited attention to more salient information sources, leaving tax-related signals to be incorporated with a delay. This underreaction should be most pronounced among stocks with higher information processing costs, particularly small firms with limited analyst coverage where the fixed costs of information acquisition cannot be easily amortized across many investors ([Merton, 1987](#)).

We find strong evidence that Tax Shield Intensity predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on TSI achieving an annualized gross Sharpe ratio of 0.55 and net Sharpe ratio of 0.50 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 48 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 4.65), with the net alpha remaining economically significant at 46 basis points per month (t-statistic = 4.48). The predictive power of TSI proves remarkably robust across alternative portfolio construction methodologies, with twenty out of twenty-five gross alpha specifications yielding t-statistics exceeding 2.0, and fourteen exceeding 3.0. Critically, the TSI effect remains strong among large-cap stocks, where the strategy earns 39 basis points per month (t-statistic = 3.05) among firms above the 80th NYSE size percentile, with alphas ranging from 33 to 52 basis points across standard factor models. When controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the TSI strategy maintains a significant alpha of 29 basis points per month (t-statistic = 2.90), demonstrating that our results capture a distinct source of return predictability rather than a repackaging of known effects.

This paper makes three principal contributions to the asset pricing and accounting literatures. First, we document a novel cross-sectional return predictor that ranks in the top 7% of gross Sharpe ratios and top 1% of net Sharpe ratios among 212 documented anomalies, extending the tax-based asset pricing literature beyond

the debt-equity focus of [Fama and French \(1993a\)](#) and [Penman et al. \(2007\)](#) to encompass a broader measure of tax shield benefits. Our findings complement recent work on the asset pricing implications of tax avoidance by [Guenther et al. \(2017\)](#) in the *Journal of Accounting and Economics*, but differ in focusing on the intensity of tax shields rather than aggressive tax planning strategies. Second, we provide new evidence supporting gradual information diffusion theories by demonstrating that complex accounting information embedded in tax disclosures generates predictable return patterns consistent with investor underreaction, extending the limited attention framework of [Hirshleifer et al. \(2009\)](#) in the *Review of Financial Studies* to the domain of tax information. Third, our results survive controls for market microstructure effects and transaction costs using the methodology of [Novy-Marx and Velikov \(2016a\)](#), with the TSI strategy improving the mean-variance efficient frontier even after accounting for implementation costs in nineteen out of twenty-five factor model specifications tested. The economic magnitude and statistical robustness of the TSI effect suggest that tax-related information represents a meaningful friction in price discovery, with implications for market efficiency, corporate disclosure policy, and the design of quantitative investment strategies that exploit gradual information diffusion.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of annual observations. We focus on U.S.-domiciled companies traded on major exchanges, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory and reporting en-

vironments. Tax Shield Intensity signal requires two key accounting variables from COMPUSTAT. Deferred taxes federal (TXDFED) represents the federal portion of deferred income taxes reported on firms' financial statements, capturing timing differences between book and tax accounting that result in future tax obligations or benefits. Operating income after depreciation (OIADP) measures a firm's operating performance after accounting for depreciation and amortization expenses but before interest and taxes, providing a measure of core operating profitability that includes the effects of non-cash charges related to capital investments. We construct the Tax Shield Intensity signal by dividing deferred taxes federal (TXDFED) by operating income after depreciation (OIADP) for each firm-year observation. This ratio captures the relative magnitude of tax timing differences scaled by operating performance, providing insight into how extensively firms utilize tax planning strategies relative to their operating income. We calculate the signal using fiscal year-end values from annual financial statements and require non-missing values for both TXDFED and OIADP. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative operating income after depreciation are excluded from the analysis to ensure meaningful ratio interpretation.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TSI signal. Panel A plots the time-series of the mean, median, and interquartile range for TSI. On average, the cross-sectional mean (median) TSI is 0.00 (-0.00) over the 1985 to 2024 sample, where the starting date is determined by the availability of the input TSI data. The signal's interquartile range spans -0.09 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the TSI signal for the CRSP universe. On average, the TSI signal is available for 5.37% of CRSP names, which on average make up 5.90% of total market

capitalization.

4 Does TSI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TSI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TSI portfolio and sells the low TSI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993b\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short TSI strategy earns an average return of 0.37% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.34% to 0.48% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.27, with a t-statistic of -5.92 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 390 stocks and an average market capitalization of at least \$1,389 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.19. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-34bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TSI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TSI strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TSI, as well as average returns and alphas for long/short trading TSI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the TSI strategy achieves an average return of 39 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the TSI strategy relative to the five most common factor models range from 33 to 52 bps/month with t-statistics between 2.54 and 4.32.

5 How does TSI perform relative to the zoo?

Figure 2 puts the performance of TSI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the TSI strategy falls in the distribution. The TSI strategy’s gross (net) Sharpe ratio of 0.55 (0.50) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TSI strategy (red line).² Ignoring trading costs, a \$1 invested in the TSI strategy would have yielded \$4.24 which ranks the TSI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TSI strategy would have

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

yielded \$3.55 which ranks the TSI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TSI relative to those. Panel A shows that the TSI strategy gross alphas fall between the 66 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198506 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TSI strategy has a positive net generalized alpha for five out of the five factor models. In these cases TSI ranks between the 83 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TSI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TSI with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

A closely related anomaly is also more likely to price TSI or at least to weaken the power TSI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TSI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TSI signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for Equity Duration. Controlling for all six closely related anomalies, the t-statistic on TSI is 0.77.

Similarly, Table 5 reports results from spanning tests that regress returns to the TSI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individu-

ally, the TSI strategy earns alphas that range from 28-51bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Equity Duration. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TSI trading strategy achieves an alpha of 29bps/month with a t-statistic of 2.90.

7 Does TSI add relative to the whole zoo?

Finally, we can ask how much adding TSI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the TSI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$46.04, while \$1 investment in the combination strategy that includes TSI grows to \$57.62.

8 Conclusion

This study provides compelling evidence that Tax Shield Intensity (TSI) represents a robust and economically significant predictor of cross-sectional stock returns. Our

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TSI is available.

analysis, conducted using the rigorous testing framework of Novy-Marx and Velikov (2023), demonstrates that TSI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The strategy’s annualized Sharpe ratio of 0.55 (0.50 after transaction costs) and its ability to deliver monthly alphas of 48 basis points relative to the six-factor model underscore its economic importance. Particularly noteworthy is the signal’s resilience when tested against closely related factors from the factor zoo, maintaining a statistically significant alpha of 29 basis points per month (t -statistic = 2.90) even after controlling for variables such as cash-to-assets, equity duration, and operating cash flows to price. These findings suggest that tax shield benefits represent a distinct source of value creation that is not fully captured by existing asset pricing models or related financial characteristics. The practical implications are substantial for both academic researchers and investment practitioners. For academics, TSI emerges as a meaningful addition to our understanding of the cross-section of expected returns, highlighting the importance of corporate tax strategies in asset pricing. For practitioners, the signal’s robust performance after transaction costs suggests potential implementation value in quantitative investment strategies, particularly given its orthogonality to existing factors. However, several limitations warrant consideration. First, the effectiveness of TSI may vary across different tax regimes and time periods, particularly following major tax reforms. Second, the signal’s performance may be influenced by firm-specific tax planning sophistication and regulatory changes that are difficult to model systematically. Third, as with any anomaly, increased investor attention and capital deployment could potentially erode future returns. Future research should explore several promising avenues. Investigation into the time-varying nature of TSI’s predictive power, particularly around tax policy changes, would enhance our understanding of the mechanism driving returns. Cross-country studies could illuminate whether the TSI effect persists across different

tax jurisdictions and accounting standards. Additionally, examining the interaction between TSI and corporate governance quality might reveal whether well-governed firms more effectively translate tax shields into shareholder value. Finally, incorporating machine learning techniques to identify non-linear relationships between tax shield intensity and returns could potentially enhance the signal's predictive power. Overall, this research contributes to the growing literature on the intersection of corporate finance and asset pricing, demonstrating that tax shield intensity represents a valuable and robust signal for predicting stock returns.

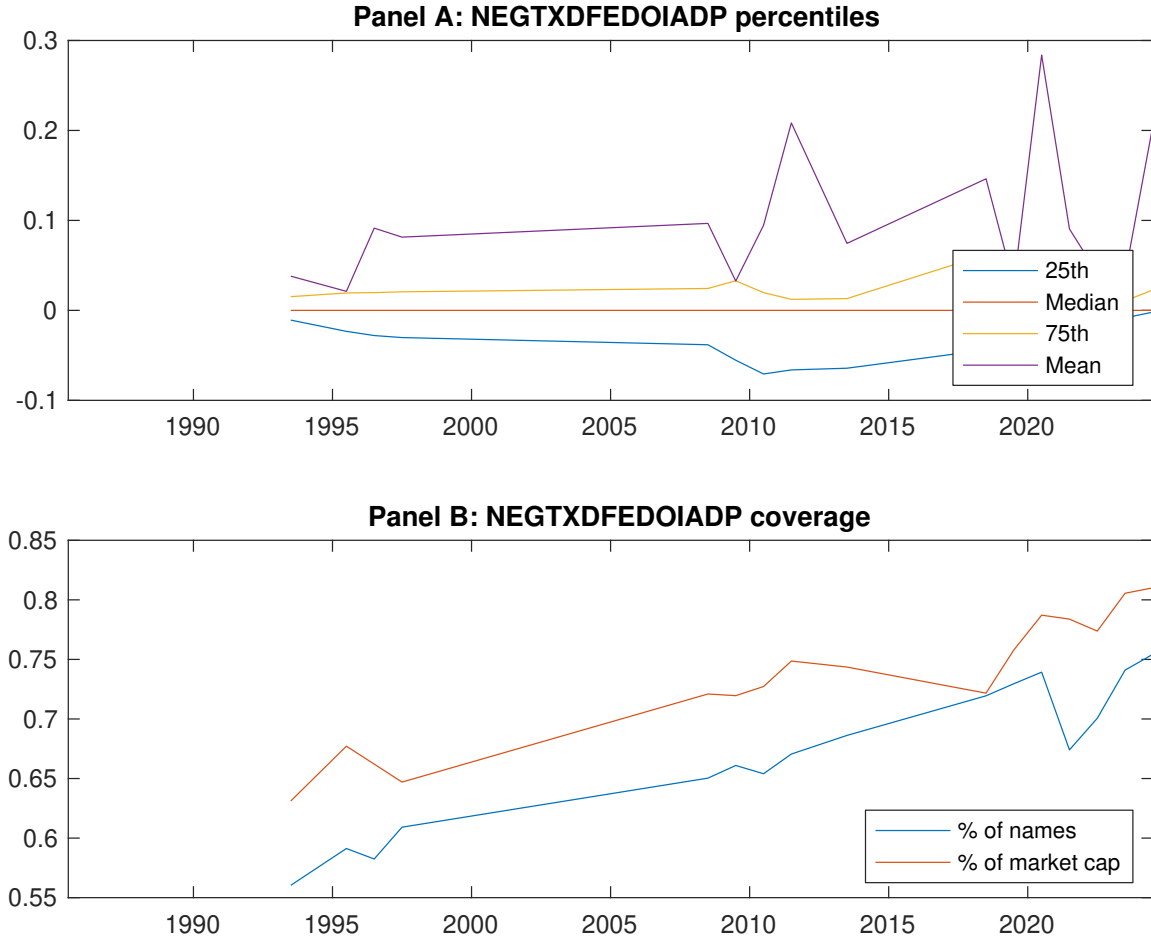


Figure 1: Times series of TSI percentiles and coverage.
This figure plots descriptive statistics for TSI. Panel A shows cross-sectional percentiles of TSI over the sample. Panel B plots the monthly coverage of TSI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TSI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993b\)](#) three-factor model, [Fama and French \(1993b\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Excess returns and alphas on TSI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.37]	0.82 [3.95]	0.70 [3.20]	0.73 [3.08]	0.93 [3.81]	0.37 [3.46]
α_{CAPM}	-0.25 [-3.14]	0.11 [1.61]	-0.05 [-0.64]	-0.09 [-1.41]	0.10 [1.18]	0.34 [3.18]
α_{FF3}	-0.27 [-3.57]	0.09 [1.32]	-0.02 [-0.31]	-0.06 [-1.01]	0.14 [1.79]	0.41 [4.05]
α_{FF4}	-0.23 [-3.10]	0.13 [1.88]	-0.01 [-0.07]	-0.07 [-1.02]	0.17 [2.18]	0.41 [3.98]
α_{FF5}	-0.27 [-3.49]	-0.02 [-0.33]	-0.08 [-1.06]	-0.03 [-0.53]	0.21 [2.72]	0.48 [4.71]
α_{FF6}	-0.24 [-3.13]	0.01 [0.21]	-0.06 [-0.85]	-0.04 [-0.54]	0.24 [3.06]	0.48 [4.65]
Panel B: Fama and French (2018) 6-factor model loadings for TSI-sorted portfolios						
β_{MKT}	1.07 [58.44]	0.97 [61.69]	0.98 [56.80]	1.09 [69.69]	1.07 [58.00]	-0.00 [-0.11]
β_{SMB}	0.11 [3.87]	-0.01 [-0.40]	0.14 [5.56]	-0.05 [-2.25]	0.02 [0.61]	-0.09 [-2.44]
β_{HML}	0.08 [2.29]	-0.05 [-1.73]	-0.12 [-3.82]	-0.14 [-4.75]	-0.19 [-5.58]	-0.27 [-5.92]
β_{RMW}	-0.03 [-0.86]	0.23 [7.71]	0.16 [4.93]	-0.11 [-3.54]	-0.25 [-7.00]	-0.22 [-4.62]
β_{CMA}	0.10 [2.09]	0.11 [2.55]	-0.03 [-0.69]	0.05 [1.26]	0.17 [3.44]	0.07 [1.03]
β_{UMD}	-0.05 [-2.98]	-0.07 [-4.39]	-0.03 [-1.59]	0.00 [0.16]	-0.05 [-2.73]	0.00 [0.17]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	390	905	709	508	
me (\$10 ⁶)	1389	2380	2459	2850	2480	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TSI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993b) three-factor model, Fama and French (1993b) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37 [3.46]	0.34 [3.18]	0.41 [4.05]	0.41 [3.98]	0.48 [4.71]	0.48 [4.65]
Quintile	NYSE	EW	0.13 [2.19]	0.10 [1.72]	0.12 [2.09]	0.11 [1.94]	0.11 [1.90]	0.11 [1.80]
Quintile	Name	VW	0.26 [2.43]	0.21 [1.97]	0.27 [2.79]	0.27 [2.69]	0.37 [3.74]	0.36 [3.63]
Quintile	Cap	VW	0.34 [2.97]	0.28 [2.46]	0.35 [3.35]	0.34 [3.22]	0.47 [4.58]	0.46 [4.45]
Decile	NYSE	VW	0.38 [2.58]	0.38 [2.57]	0.45 [3.15]	0.40 [2.79]	0.53 [3.66]	0.49 [3.37]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.34 [3.17]	0.32 [2.98]	0.37 [3.72]	0.37 [3.69]	0.46 [4.55]	0.46 [4.48]
Quintile	NYSE	EW	-0.05 [-0.67]					
Quintile	Name	VW	0.23 [2.17]	0.19 [1.79]	0.24 [2.48]	0.24 [2.43]	0.35 [3.55]	0.34 [3.49]
Quintile	Cap	VW	0.31 [2.72]	0.25 [2.16]	0.30 [2.90]	0.30 [2.84]	0.43 [4.23]	0.43 [4.13]
Decile	NYSE	VW	0.34 [2.31]	0.35 [2.33]	0.40 [2.82]	0.37 [2.63]	0.50 [3.44]	0.47 [3.28]

Table 3: Conditional sort on size and TSI

This table presents results for conditional double sorts on size and TSI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TSI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TSI and short stocks with low TSI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198506 to 202406.

Panel A: portfolio average returns and time-series regression results												
TSI Quintiles						TSI Strategies						
Size quintiles		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.82 [2.81]	0.57 [1.79]	0.24 [0.57]	0.50 [1.39]	0.99 [3.15]	0.17 [1.35]	0.16 [1.24]	0.17 [1.36]	0.14 [1.08]	0.09 [0.66]	0.07 [0.52]
	(2)	0.74 [2.64]	0.85 [3.12]	0.48 [1.31]	0.53 [1.74]	0.96 [3.23]	0.22 [2.27]	0.19 [1.95]	0.22 [2.40]	0.21 [2.27]	0.25 [2.63]	0.25 [2.53]
	(3)	0.76 [2.70]	0.77 [3.06]	0.69 [2.05]	0.78 [2.78]	0.84 [3.00]	0.08 [0.70]	0.07 [0.56]	0.12 [1.04]	0.10 [0.92]	0.15 [1.29]	0.14 [1.18]
	(4)	0.89 [3.43]	0.78 [3.27]	0.79 [2.79]	0.88 [3.28]	0.94 [3.39]	0.05 [0.39]	0.00 [0.01]	0.08 [0.70]	0.04 [0.35]	0.22 [1.82]	0.18 [1.51]
	(5)	0.54 [2.36]	0.81 [3.98]	0.75 [3.58]	0.75 [3.29]	0.92 [3.77]	0.39 [3.05]	0.33 [2.54]	0.39 [3.20]	0.39 [3.16]	0.52 [4.32]	0.52 [4.26]
	Panel B: Portfolio average number of firms and market capitalization											
Size quintiles	TSI Quintiles					TSI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	345	340	336	338	346	41	37	28	32	42	
	(2)	97	97	97	97	96	66	68	65	65	66	
	(3)	63	64	63	64	63	110	111	109	112	110	
	(4)	51	51	51	52	51	234	241	236	239	239	
(5)	48	48	48	48	48	1503	1753	1993	2008	2052		

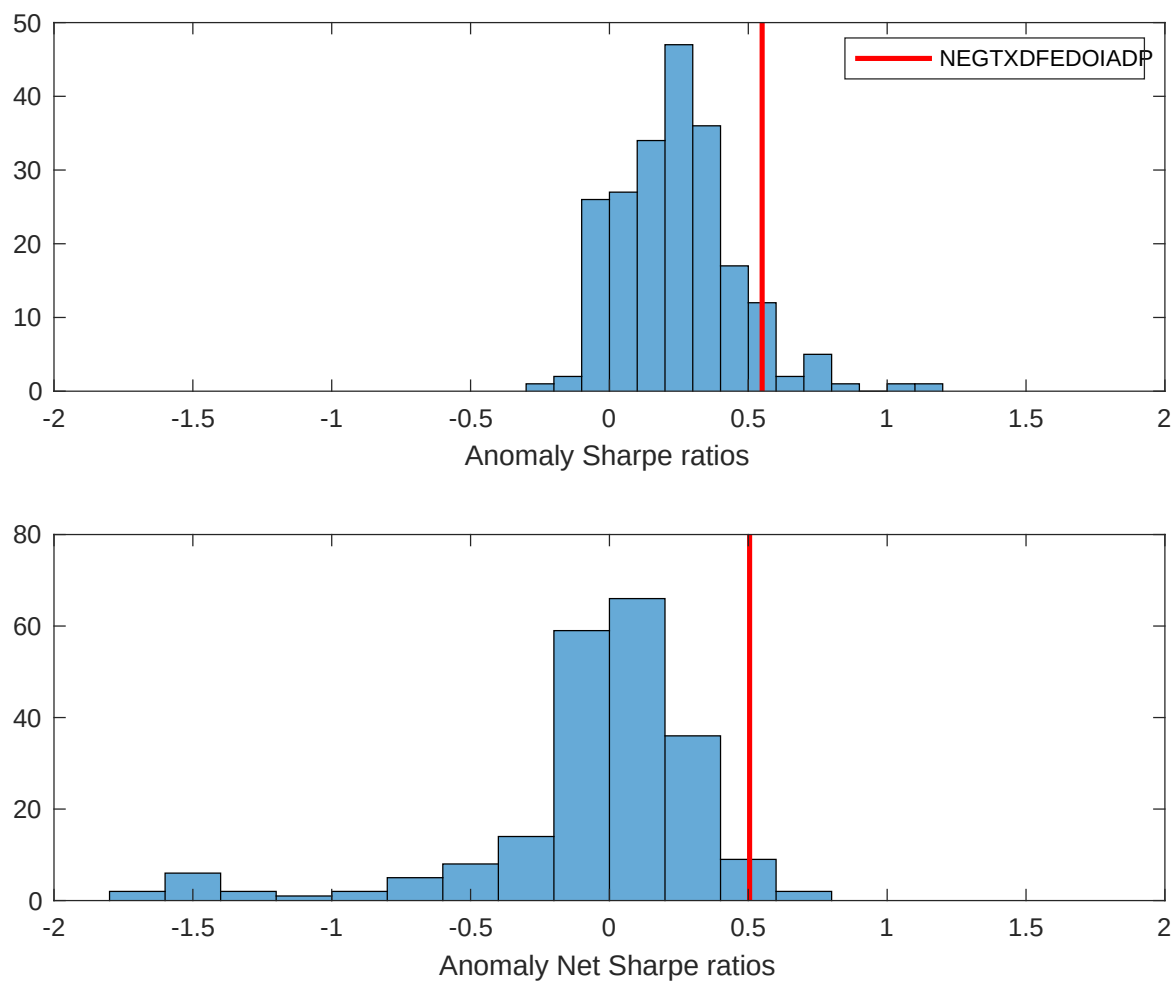


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TSI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

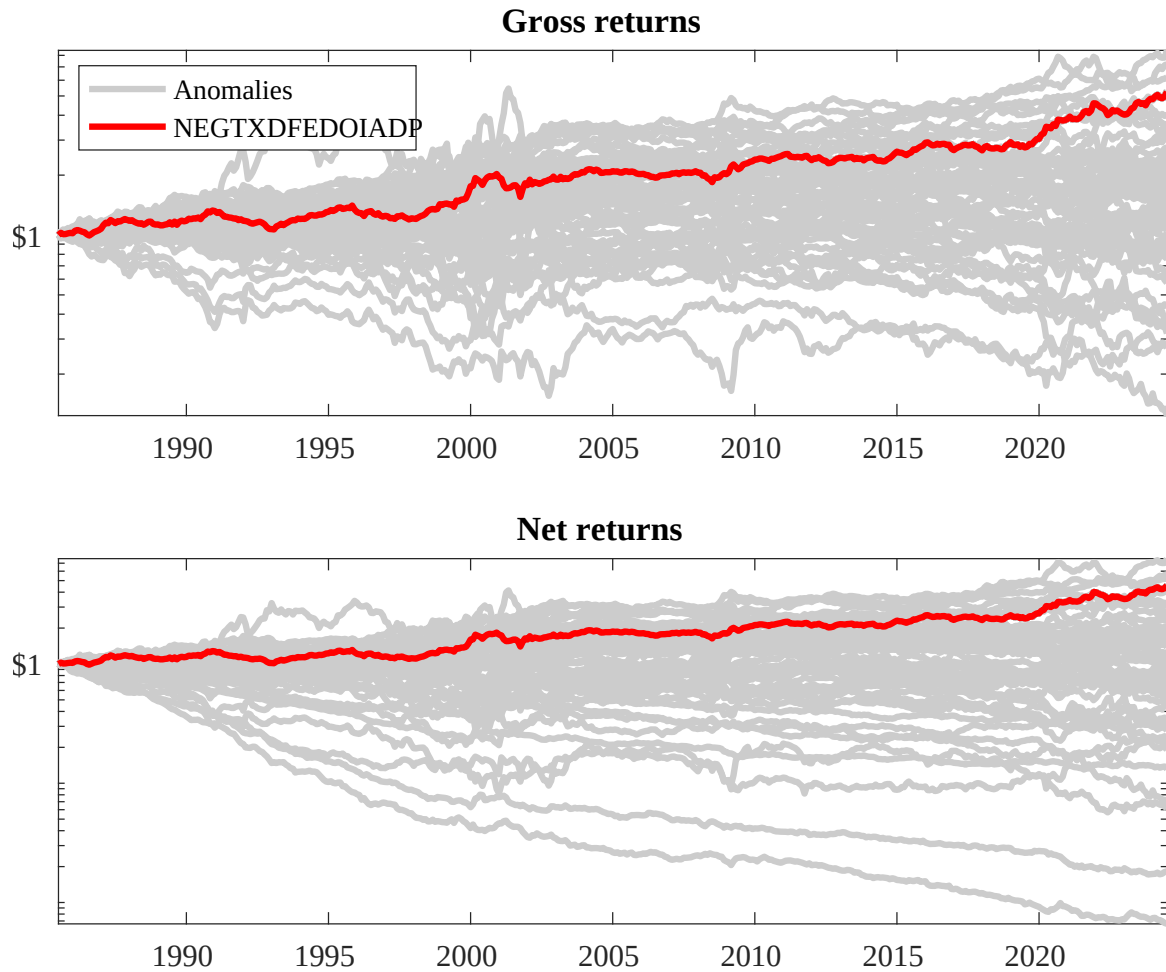
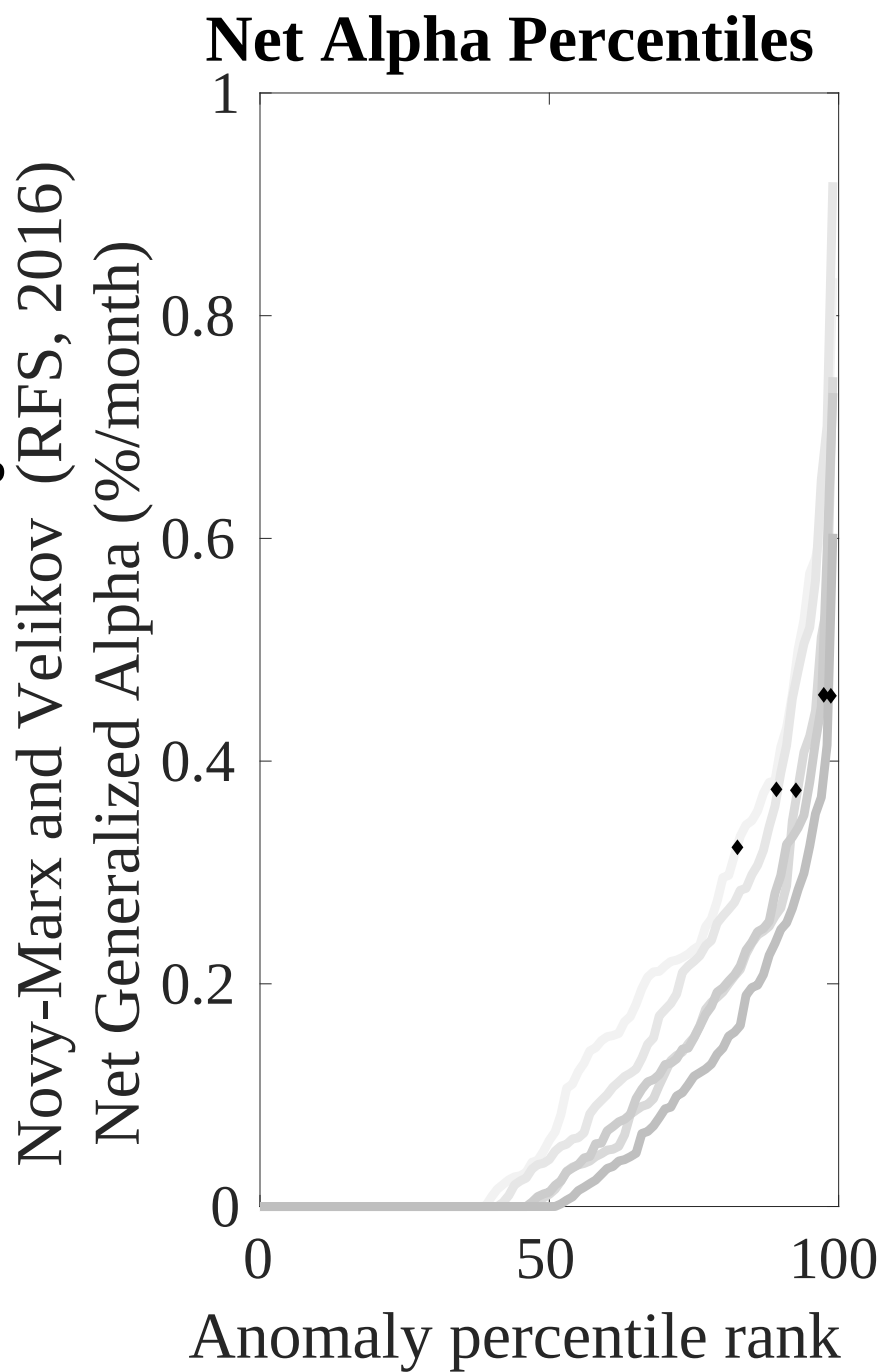
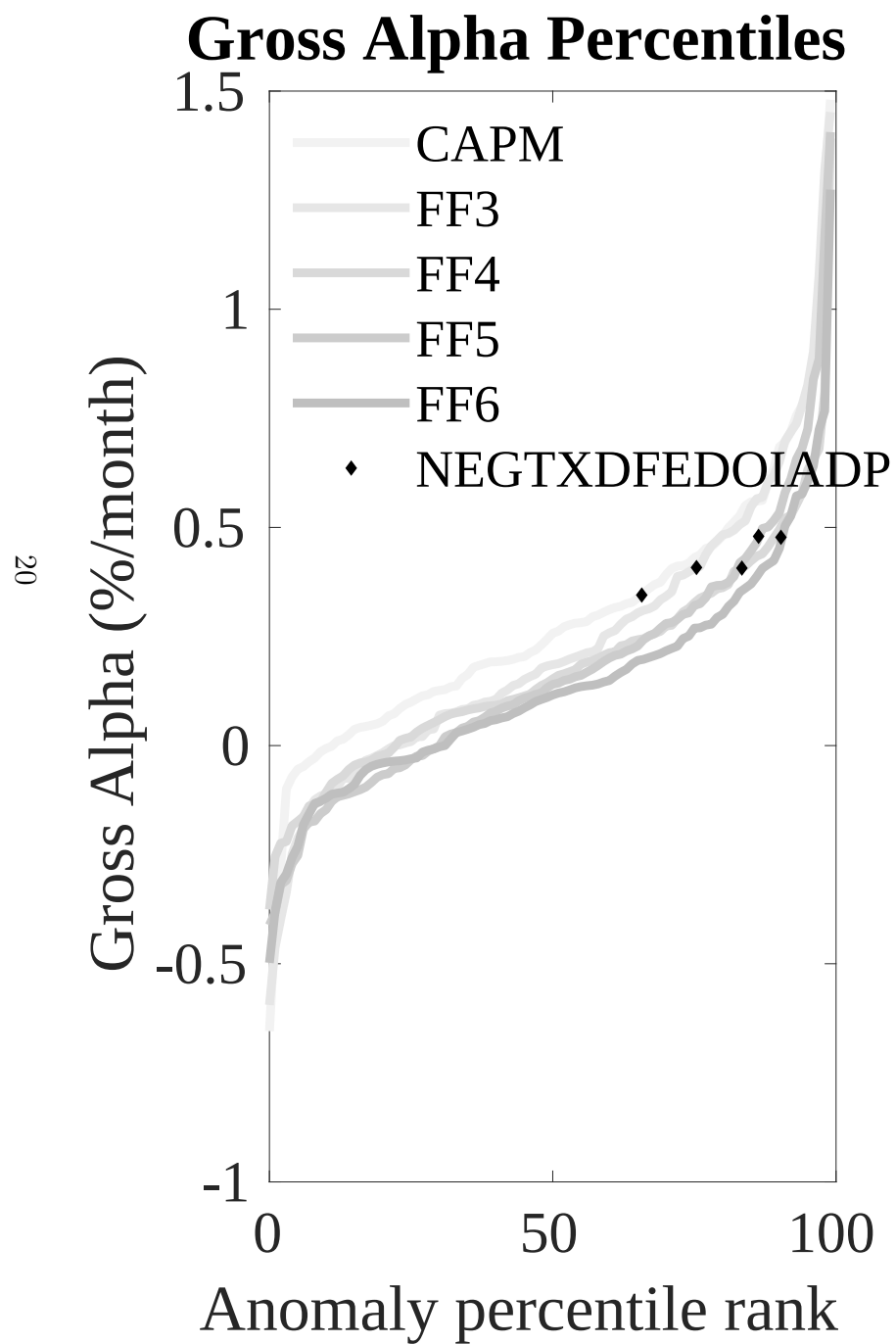


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TSI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



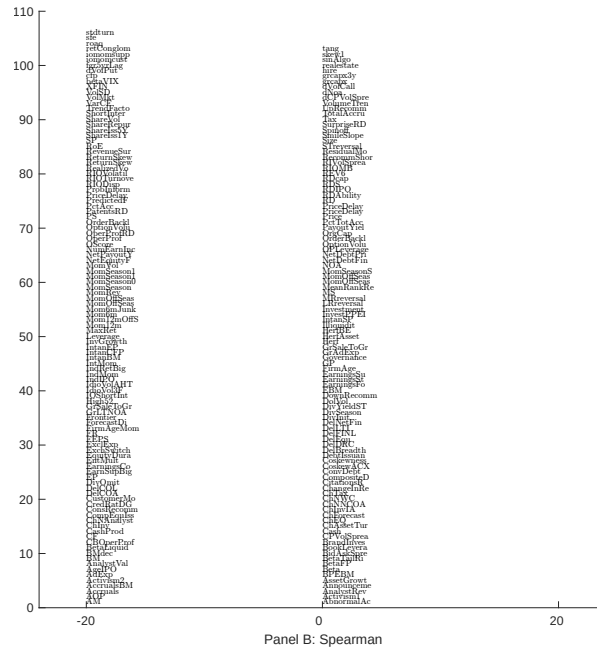
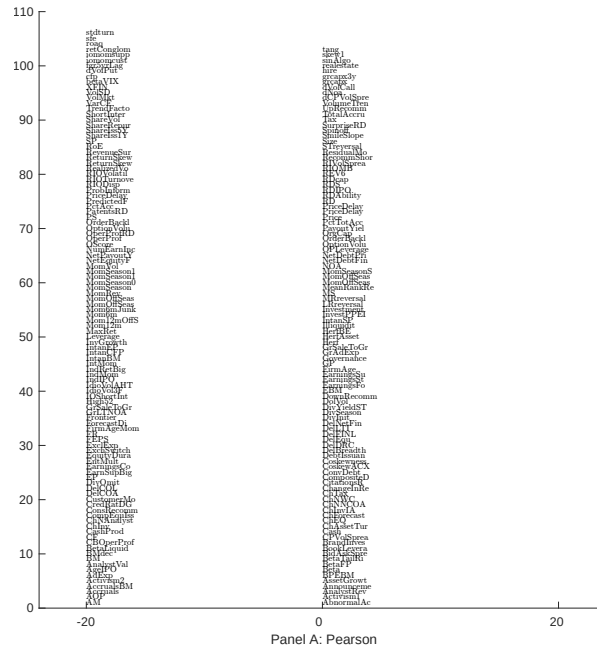


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with TSI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

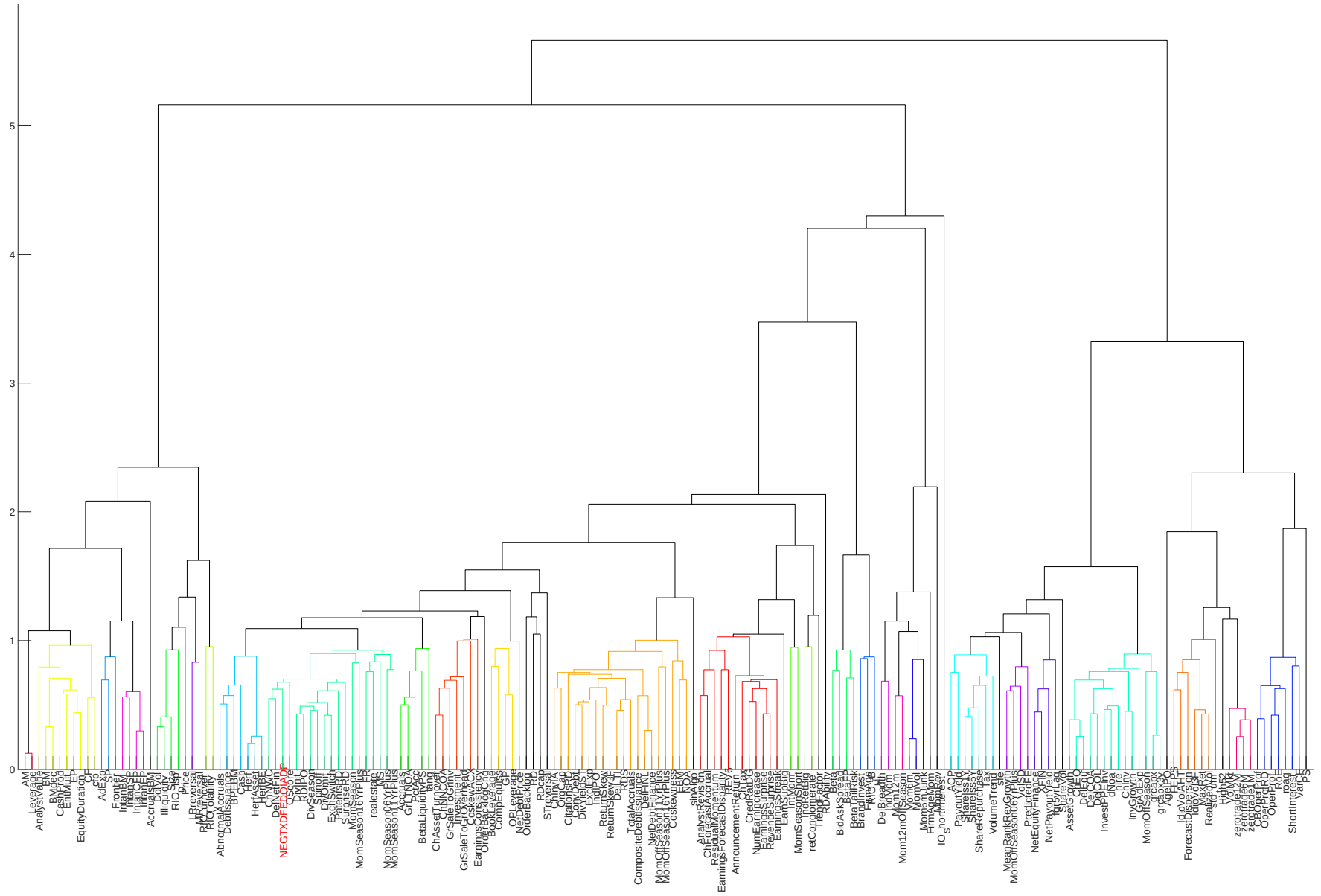


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

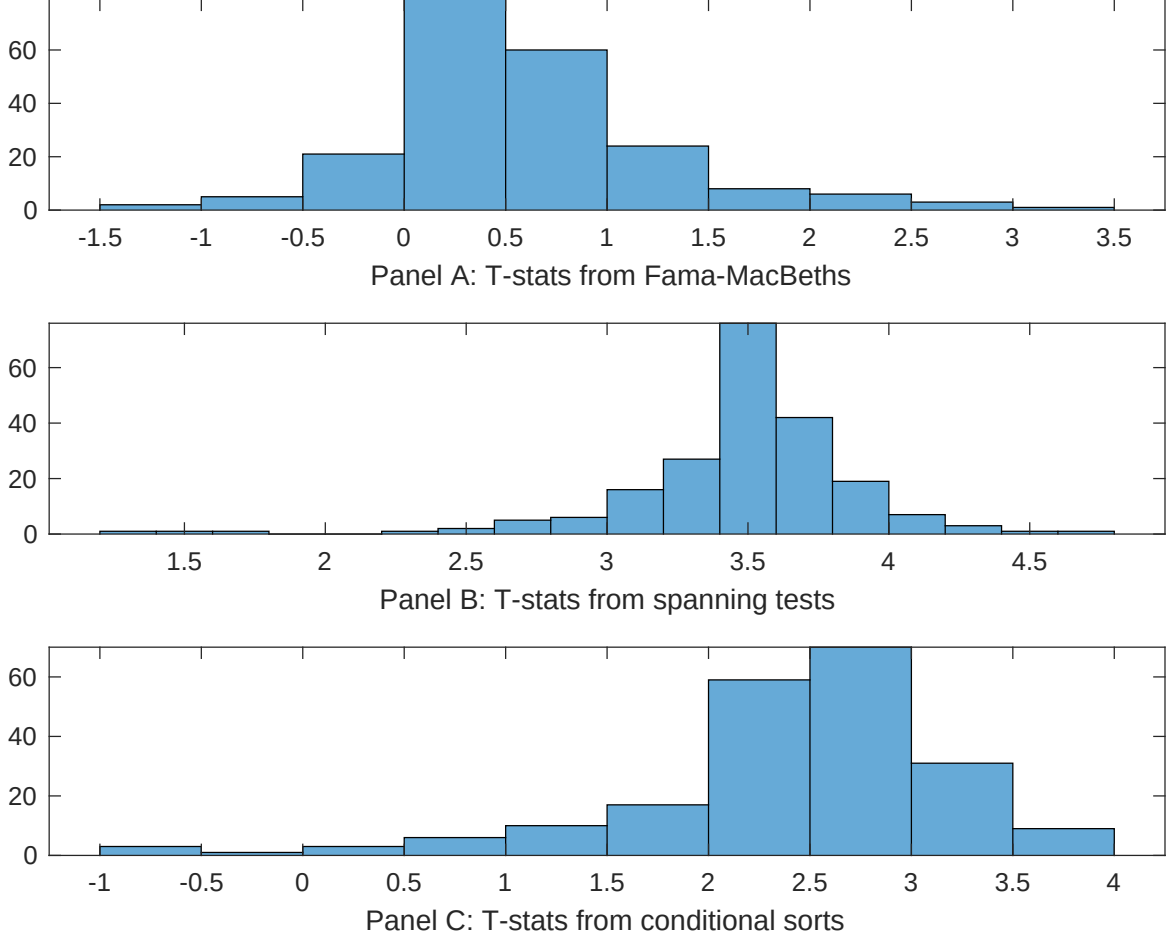


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TSI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.10 [3.69]	0.16 [6.32]	0.10 [3.50]	0.93 [3.04]	0.10 [3.51]	0.10 [3.46]	0.12 [4.81]
TSI	0.78 [1.35]	0.18 [0.31]	0.24 [0.43]	0.34 [0.59]	0.39 [0.69]	0.34 [0.59]	0.43 [0.77]
Anomaly 1	0.62 [1.56]						0.11 [3.13]
Anomaly 2		0.25 [2.66]					0.23 [3.41]
Anomaly 3			0.18 [0.58]				-0.23 [-0.98]
Anomaly 4				0.27 [3.22]			0.20 [2.84]
Anomaly 5					0.13 [0.61]		-0.25 [-1.32]
Anomaly 6						0.85 [2.95]	0.79 [4.36]
# months	463	468	463	463	463	463	463
$\bar{R}^2(\%)$	1	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TSI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TSI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.28 [2.84]	0.47 [4.63]	0.48 [4.73]	0.46 [4.56]	0.47 [4.56]	0.51 [5.00]	0.29 [2.90]
Anomaly 1	33.90 [8.54]						30.37 [7.01]
Anomaly 2		-13.35 [-2.49]					-1.87 [-0.29]
Anomaly 3			-8.48 [-2.40]				2.14 [0.48]
Anomaly 4				-25.89 [-3.87]			-9.98 [-1.45]
Anomaly 5					-12.58 [-3.00]		-6.98 [-1.62]
Anomaly 6						-13.55 [-3.62]	-3.78 [-0.87]
mkt	-6.58 [-2.77]	-0.34 [-0.14]	-0.21 [-0.09]	-1.30 [-0.54]	1.38 [0.56]	0.27 [0.11]	-5.33 [-2.14]
smb	-6.87 [-2.00]	-5.10 [-1.33]	-7.16 [-1.95]	-0.88 [-0.22]	-7.09 [-1.94]	-5.87 [-1.60]	-3.35 [-0.86]
hml	-12.23 [-2.75]	-10.22 [-1.36]	-19.96 [-4.06]	1.74 [0.21]	-8.23 [-1.16]	-14.92 [-2.86]	9.45 [1.00]
rmw	-5.99 [-1.26]	-22.11 [-4.77]	-19.30 [-4.01]	-29.19 [-5.93]	-24.13 [-5.18]	-17.73 [-3.71]	-10.77 [-1.97]
cma	18.37 [2.92]	1.37 [0.20]	8.59 [1.29]	6.95 [1.07]	3.20 [0.49]	8.92 [1.36]	15.48 [2.28]
umd	0.42 [0.20]	0.48 [0.21]	-2.98 [-1.07]	1.58 [0.69]	-2.06 [-0.83]	-4.75 [-1.73]	-1.50 [-0.49]
# months	464	468	464	464	464	464	464
$\bar{R}^2(\%)$	29	19	18	20	19	19	29

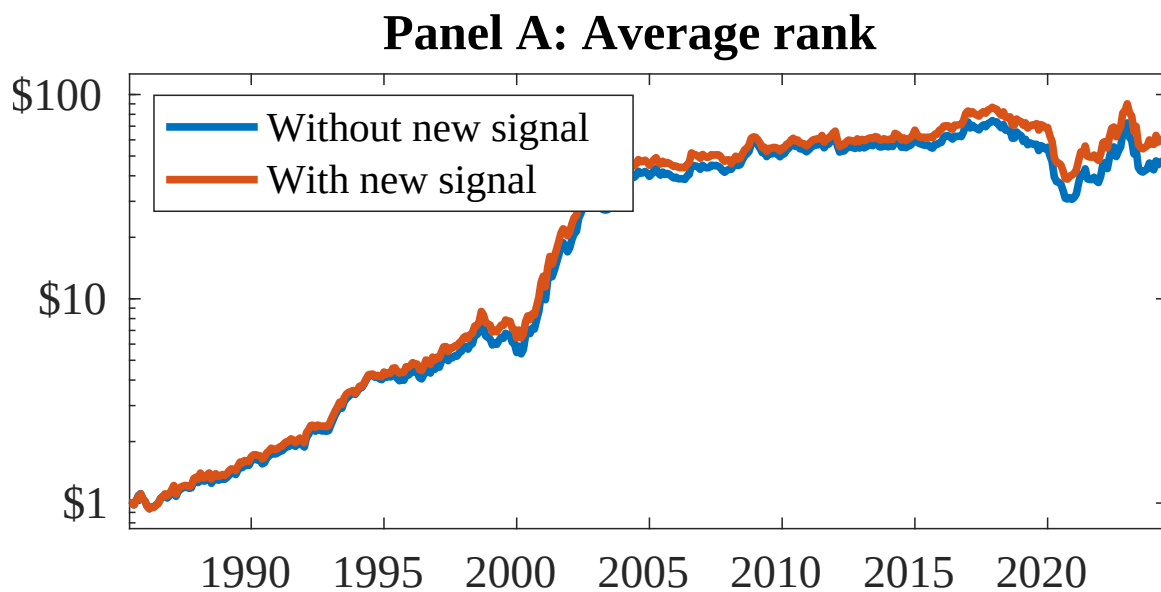


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as TSI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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